SET NO. 24

PRODUCT INVENTORY MANAGEMENT

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Course:DSA0601 Data Handling and Visualization

Dataset: Product Inventory

Product ID	Product Name	Quantity Available
1	Product A	250
2	Product B	175
3	Product C	300
4	Product D	200
5	Product E	220

Question 1: Inventory Quantity Bar Chart

Aim:

To visualize the quantity available for each product using a bar chart.

Algorithm:

1. Store product names and quantities.

2. Create a bar chart using barplot().

3. Label the x-axis and y-axis.

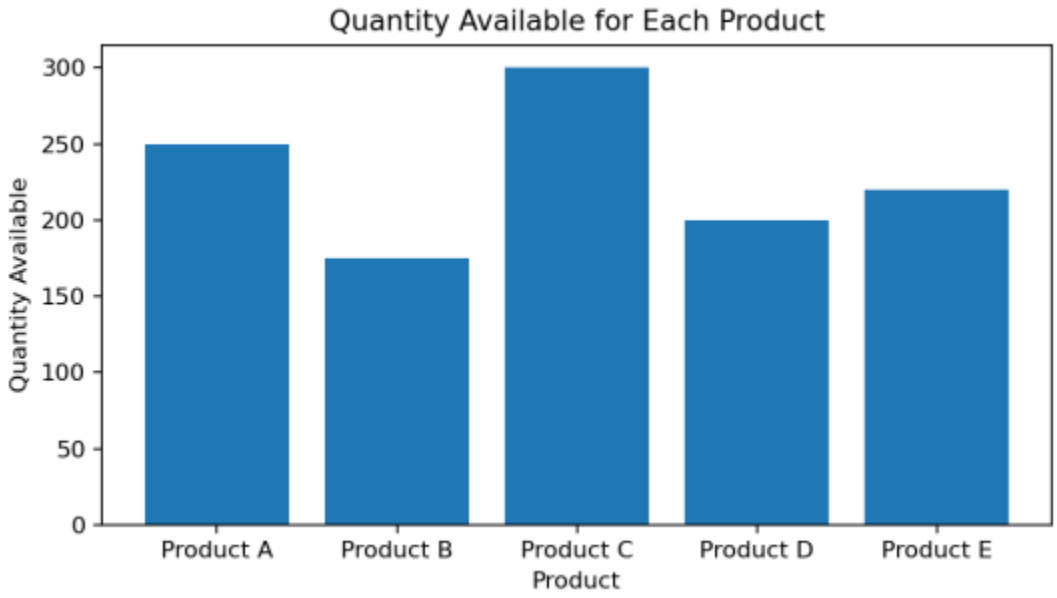
4. Add a title and display the chart.

Code (R):

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
quantity <- c(250, 175, 300, 200, 220)

barplot(quantity, names.arg=product,
        main="Quantity Available for Each Product",
        xlab="Product", ylab="Quantity Available",
        col="skyblue")
```

Output: A bar chart comparing the available quantity of Products A to E.



Result: Product C has the highest quantity (300), while Product B has the lowest quantity (175).

Question 2: Stacked Bar Chart by Product Category

Aim: To create a stacked bar chart showing the quantity of each product divided across categories.

Algorithm:

1. Store product names and category-wise quantities.

2. Draw the first bar series.

3. Add the second series using the bottom argument.

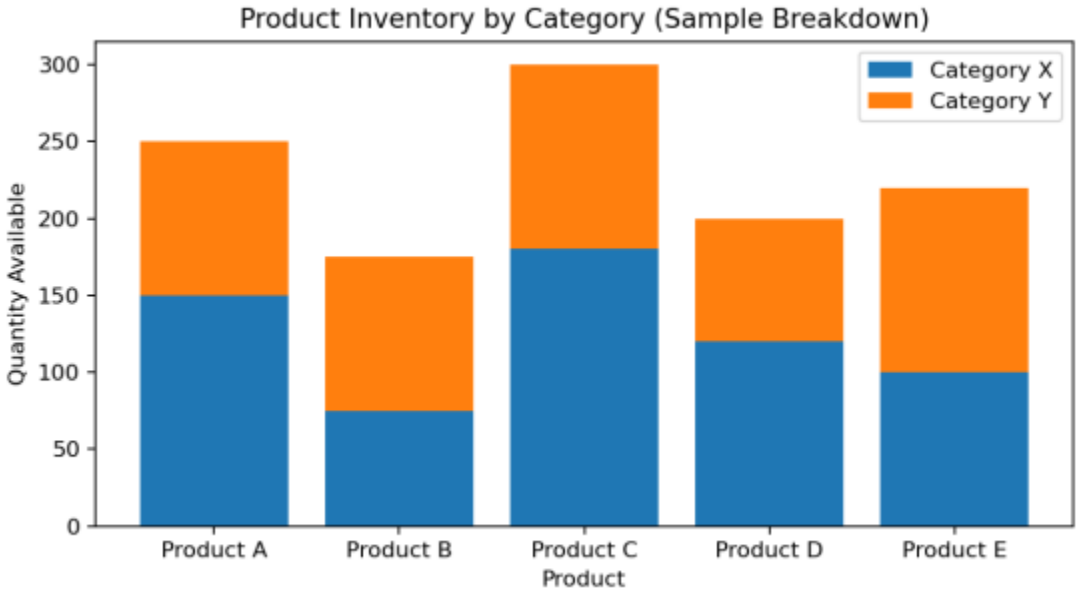
4. Add a legend, labels, and title.

Code (R):

```
product <- c("Product A", "Product B", "Product C", "Product D", "Product E")
category_x <- c(150, 75, 180, 120, 100)
category_y <- c(100, 100, 120, 80, 120)

barplot(rbind(category_x, category_y), beside=FALSE,
        names.arg=product,
        col=c("skyblue", "orange"),
        main="Product Inventory by Category",
        xlab="Product", ylab="Quantity Available")
legend("topleft", legend=c("Category X", "Category Y"),
       fill=c("skyblue", "orange"))
```

Output: A stacked bar chart showing each product's total quantity divided into two sample categories.



Result: The chart demonstrates how category-wise quantities can be compared. Replace sample category values with actual category data.

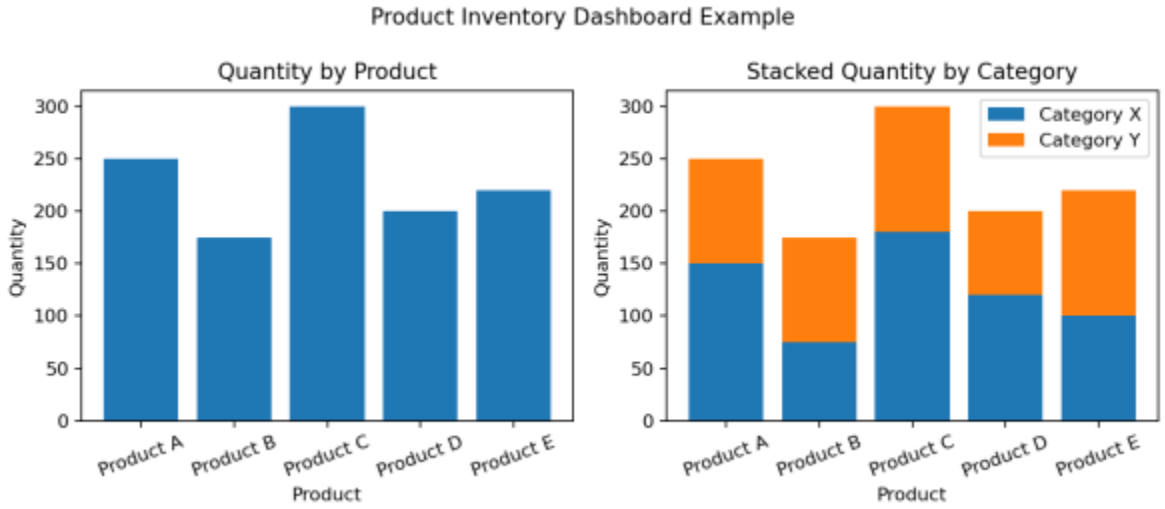
Question 3: Tableau Inventory Dashboard

Aim: To build a dashboard combining a product quantity bar chart and a category-wise stacked bar chart.

Algorithm:

1. Connect the inventory data in Tableau.
2. Create a bar chart using Product Name and Quantity Available.
3. Create a stacked bar chart using Product Name, Category, and Quantity.
4. Add both worksheets to a dashboard.
5. Add a title and filters for interactive exploration.

Output: A dashboard example combining inventory quantity and stacked category views.



Result: The dashboard supports quick comparison of product stock and category contributions. The category example uses assumed data.

Question 4: Price and Quantity Scatter Plot

Aim: To explore the relationship between product price and quantity available using a scatter plot.

Algorithm: 1. Store product prices and quantities.

2. Plot price on the x-axis and quantity on the y-axis.

3. Label the axes and add a title.

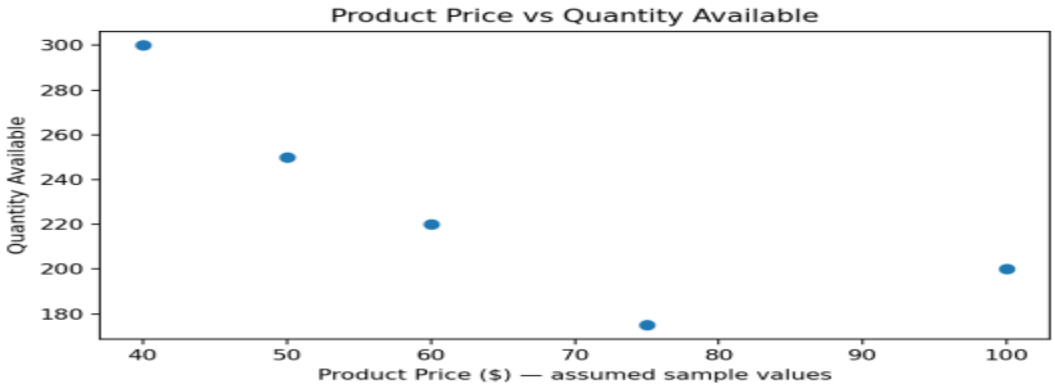
4. Observe the point pattern.

Code (R):

```
price <- c(50, 75, 40, 100, 60)  # assumed sample prices
quantity <- c(250, 175, 300, 200, 220)

plot(price, quantity, pch=19, col="blue",
    main="Product Price vs Quantity Available",
    xlab="Product Price ($)",
    ylab="Quantity Available")
```

Output: A scatter plot showing the assumed product prices



Result: The scatter plot is useful for exploring whether price and inventory quantity are related when real price data are available.